

TEAM 31

APPLIED AI & DATA SCIENCE FOR THE FIRE DEPARTMENT

SOFTWARE QUALITY PLAN

3/21/26

OVERVIEW

1. Project Description

As the City of Scottsdale responds to incidents such as fires, rescues, collisions, and electrical hazards, thousands of incident reports are generated which contain valuable information about the fire department's response readiness. While this information is cataloged and stored for internal reference, there was a need for a system that could transform this data into meaningful, actionable insights.

Team 31 was tasked with developing a software solution to address this need and assist the Scottsdale Fire Department in identifying areas for improvement and enhancing response efficiency. Team 31 consists of five members (listed alphabetically): Zachary Alexander, Damion Dray, Michael Krasnik, Leroy Sturdivent, and Andrew Tellez. The team works directly with the Scottsdale Fire Department through its project sponsor, Franklin Liu.

Through close collaboration with the department, Team 31 has designed and begun development of a system aimed at addressing the growing demand for incident data analytics. This solution, called FAMAR (Fire and Medical Analytic Report), is an online dashboard system that processes KPI (Key Performance Indicator) data from Fire and EMS incidents and presents it through interactive visualizations.

FAMAR utilizes uploaded CSV and Excel files containing incident data organized according to the NFIRS (National Fire Incident Reporting System) standard. The system transforms this data into charts and dashboards, enabling users to analyze performance metrics and identify trends. It also supports PDF export functionality, allowing users to generate comprehensive reports within a selected date range. Additionally, FAMAR integrates generative AI capabilities, enabling users to query and analyze data without directly interacting with the database. The AI component can also leverage historical data to support trend prediction and provide insights into potential efficiency improvements.

The implementation of FAMAR will enable the Scottsdale Fire Department to enhance its operational effectiveness by monitoring unit performance across Fire and EMS incidents. These insights will support improved efficiency and faster response times. Ultimately, by improving these key performance metrics, the system has the potential to strengthen emergency response capabilities and contribute to saving lives.

2. High-Level Requirements

The below listed high-level requirements define the functionality, performance, security and usability that is expected by the stakeholders of the FAMAR project. These requirements are based on the initial project goal and project plan

as well as slight scope changes throughout the project documented and discussed on calls with our stakeholders. In addition to meeting all project requirements, the following high-level requirements apply:

1. The system shall allow authorized users to upload CSV or Excel files containing Fire or EMS data respectively.
2. The system shall be capable of storing and maintaining a minimum of 36 months of historical Fire / EMS data.
3. The system shall generate KPI dashboard charts as well as metrics that are derived from the data uploaded by the end user.
4. The system shall generate AI-assisted PDF reports that summarize the KPIs and charts selected for a given period.
5. The system shall restrict access to users with “viewer” access for uploading any data.
6. The system shall display the KPI dashboard charts as well as metrics within 1000 ms of the selected filters.
7. The system shall allow the end user to add, remove, or reposition the KPI charts in the dashboard and save their preferences to their profile.
8. The system shall allow the user to monitor the Fire data and the EMS data separately in their respective viewport.
9. The system shall display KPI dashboard charts in a revolving manner when selected by the end user with the ability to filter data and alter display duration times to their own discretion.
10. The system shall include an AI companion to help navigate, research, and organize data for an user experience with minimal complications.

3. Quality Metrics

This section of the quality plan provides a description of the metrics and acceptance level for each quality measurement. The table below gives the metric, metric description, acceptance criteria, and the related requirements for each metric.

Metric	Metric Description	Target / Acceptance Level	Related Requirement(s)
Upload Validation Success Rate	Percentage of valid Fire and EMS CSV or Excel files that are accepted, validated, and written to the database without ingestion errors.	100% of approved-format test files upload successfully.	R1
Rejected File Blocking Rate	Percentage of invalid file types or malformed uploads that are rejected with an error before any data is stored.	100% of invalid-format or malformed test files are rejected.	R1
Historical Data Retention Span	Length of historical Fire and EMS data that can be stored and queried in the production schema without loss of access.	At least 36 months of uploaded history are retained and queryable.	R2

Historical Record Integrity	Percentage of uploaded incident, unit response, and EMS records that remain queryable after storage and reload operations.	At least 99.5% of imported records are preserved with matching identifiers and timestamps.	R2
KPI Generation Coverage	Percentage of required KPI dashboard services that return valid chart or metric data for a selected date range after upload.	100% of required KPI endpoints return successful responses for valid ranges.	R3
KPI Calculation Accuracy	Difference between displayed KPI values and independently calculated expected values using the same source dataset.	0 critical discrepancies and all verified KPI values within plus or minus 1% of expected results.	R3
PDF Report Generation Success Rate	Percentage of report-generation attempts that produce a downloadable PDF containing the selected charts, KPIs, and reporting period.	At least 95% successful report generation across planned report scenarios.	R4
Report Content Completeness	Percentage of selected report elements that appear correctly in the generated PDF output.	100% of selected KPIs, charts, and reporting period labels appear in the PDF.	R4
Viewer Upload Denial Rate	Percentage of upload attempts by viewer-role accounts that are blocked by role-based access controls.	100% of viewer upload attempts are denied.	R5
Role Enforcement Accuracy	Percentage of role-based UI and API checks that correctly allow analyst and admin uploads while denying viewer uploads.	100% correct enforcement for viewer, analyst, and admin test cases.	R5
Dashboard Filter Response Time	Elapsed time from applying valid dashboard filters to all KPI cards and charts rendering updated results.	95th percentile less than or equal to 1000 ms in the defined test environment.	R6
Dashboard Load Completion Rate	Percentage of filter changes that complete without client or server errors for the selected reporting range.	At least 99% successful refreshes during performance test runs.	R6

Protected Route Enforcement Rate	Percentage of attempts to access protected routes without a valid authenticated session that are redirected to the correct login or session-expired screen.	100% of unauthorized access attempts are blocked from protected views.	R7
Login Session Success Rate	Percentage of valid login sessions that successfully reach the authorized application home view.	At least 99% successful session establishment for valid credentials in test scenarios.	R7
Profile Completion Enforcement Rate	Percentage of users missing required username or phone values who are redirected to the profile completion flow before entering the dashboard.	100% of incomplete-profile users are redirected until required fields are saved.	R8
Profile Update Completion Success Rate	Percentage of profile completion submissions that successfully save required fields and permit dashboard access.	At least 95% successful completion of required profile updates in test scenarios.	R8
Verification Gate Enforcement Rate	Percentage of unverified accounts prevented from accessing protected dashboard functionality before administrative approval.	100% of unverified accounts are blocked from protected dashboard features.	R9
Post-Approval Access Success Rate	Percentage of approved accounts that can successfully proceed to the authorized application experience after verification is granted.	At least 95% of approved-user test accounts gain access without manual intervention.	R9
Admin User Management Success Rate	Percentage of administrator actions to approve users, update roles, and remove users that complete successfully through the admin console.	At least 95% successful completion across planned admin management scenarios.	R10
Admin Console Permission Accuracy	Percentage of admin-console access checks that correctly permit admins and deny non-admin roles.	100% correct access enforcement for admin and non-admin role tests.	R10

4. Testing Plan

This section describes how Team 31 will verify that FAMAR meets the quality metrics in Section 3. The overall strategy combines positive and negative testing, automated checks where they give repeatable evidence (API and data-layer behavior), and structured manual or semi-automated checks where the user interface, reporting, or timing must be observed. Upload and ingestion paths are exercised with representative NFIRS-style Fire and EMS datasets (valid, boundary-case, and intentionally invalid). KPI verification uses independent recomputation from the same stored data as an oracle. Role-based behavior is verified at both the UI and the API so a single layer cannot mask a defect. Performance for dashboard filtering is measured in a documented test environment (browser version, client hardware tier, network profile, and database size class) so “95th percentile $\leq$ 1000 ms” is interpretable and reproducible. Tests are mapped to the metrics they support; where one execution supports two metrics, both are named under Applies to.

Approved-format upload end-to-end verification

Applies to: Upload Validation Success Rate (R1).

Test description: Using analyst or admin credentials, upload a curated set of valid Fire and EMS files (CSV and XLSX) that conform to the project’s expected NFIRS-oriented layout and pass data-quality rules. For each file, confirm HTTP success from the upload API (for example POST /api/upload with data_type=fire or data_type=ems), confirm row counts and success payload, and confirm the data is visible through downstream read paths (for example KPI queries over a date range that includes uploaded incidents). Log each run as pass or fail.

Rationale: This test directly measures whether approved-format, valid uploads complete ingestion without error, which is exactly what Upload Validation Success Rate defines.

Invalid and malformed upload rejection suite

Applies to: Rejected File Blocking Rate (R1).

Test description: Attempt uploads that must be rejected before unsafe or inconsistent data is stored: wrong extension (for example .pdf, .txt), corrupted binary masquerading as CSV, empty files, files with wrong data_type for content, and files that fail parsing or configured data-quality rules. Confirm the API returns a 4xx with a clear error, and confirm via database or read APIs that no new incident rows were committed for rejected cases. Include client-side checks where applicable (for example the upload control rejects oversize or wrong-type files before submit).

Rationale: Negative testing is the most direct way to show 100% rejection of invalid input without storage, matching the metric and the template note that negative testing often verifies requirements best.

Historical span query test

Applies to: Historical Data Retention Span (R2).

Test description: On a production-like dataset (or a restored backup), select date filters that span at least 36 consecutive months of incident timestamps. Run representative KPI queries across that window (summary, call volume, heatmap, and at least one unit- or postal-based endpoint). Record any gaps, truncation, or query failures attributable to retention limits.

Rationale: The requirement is stated in terms of stored and queryable history; this test proves the system still serves analytics over the full retention window.

Import round-trip and record integrity check

Applies to: Historical Record Integrity (R2).

Test description: For a controlled upload, capture a statistical sample of primary keys (incident identifiers, unit response identifiers, and key timestamps) from the source file. After ingestion, query the database or export equivalent rows and compare identifiers and timestamps for at least 99.5% exact matches on the sample (full comparison for small files). Investigate and classify any mismatch as ingestion defect versus acceptable transformation (only if explicitly specified in project rules).

Rationale: The metric is about preservation of records through storage and reload; sampling with full verification on small sets gives measurable evidence aligned with the acceptance target.

KPI API coverage and contract test

Applies to: KPI Generation Coverage (R3).

Test description: Maintain a checklist of required KPI services exposed by the staging API (for example GET /api/incidents/summary, GET /api/incidents/response-times, GET /api/incidents/heatmap, GET /api/incidents/postal-breakdown, GET /api/incidents/call-volume, GET /api/incidents/unit-origin, GET /api/incidents/type-breakdown, GET /api/incidents/mutual-aid). For each endpoint, call it with a valid date range and supported region values, assert HTTP 200 and a non-error JSON shape (presence of expected top-level keys). Run after each significant backend change or before milestone demos.

Rationale: Coverage here means every required KPI path returns valid data for legitimate inputs; a checklist against live endpoints is objective and easy to evidence in the later quality report.

KPI golden-dataset reconciliation

Applies to: KPI Calculation Accuracy (R3).

Test description: Fix a golden subset of incident and unit-response data with known answers (for example total incidents in range, average response minutes, call-processing and travel splits). Using SQL or a trusted script outside the application, compute expected KPI values. Compare application/API outputs to the oracle. Classify discrepancies: acceptable if within $\pm 1\%$ (or within agreed rounding rules); critical if logic errors or wrong filters are indicated.

Rationale: Independent recomputation is the standard way to justify numerical accuracy; it ties the test tightly to the metric's acceptance criteria.

PDF report generation scenario matrix

Applies to: PDF Report Generation Success Rate (R4).

Test description: Define a matrix of report scenarios: multiple date ranges (short, full year, spanning uploaded data), different combinations of selected KPIs/charts, and typical user roles allowed to export. For each scenario, trigger AI-assisted PDF generation as implemented in the product, record success or failure, and measure the proportion that yields a downloadable PDF without server or client errors. Target at least 95% success across the matrix.

Rationale: This measures reliability of the report pipeline under realistic variation, which is what the success-rate metric captures.

PDF content completeness checklist

Applies to: Report Content Completeness (R4).

Test description: For each successful PDF from the scenario matrix, manually or with a PDF text-extraction tool verify that every user-selected KPI, chart, and the reporting period label appears in the output and matches the selection (no missing sections, no wrong date span). Mark each element present or absent.

Rationale: Success rate alone does not prove correct content; this test is specifically aligned with the metric that selected elements appear correctly.

Viewer-role upload denial test

Applies to: Viewer Upload Denial Rate (R5).

Test description: Authenticate as a viewer and attempt every path to upload data: direct navigation to upload view, API calls with viewer tokens (if applicable), and deep links. Confirm the UI blocks or hides upload per design and that the upload API returns 403/401 (or equivalent) for viewer. Count attempts versus denials; expect 100% denial for viewers.

Rationale: This is explicit negative testing for requirement R5 and maps directly to the denial-rate metric.

Role enforcement matrix (viewer, analyst, admin)

Applies to: Role Enforcement Accuracy (R5).

Test description: Build a small matrix of actions (view dashboard, change filters, upload Fire, upload EMS, access admin-only screens if present). For each role (viewer, analyst, admin), execute each action and record allowed versus denied against the specification. Include both UI behavior (for example Home routing and NavBar upload visibility) and API behavior where roles are enforced server-side.

Rationale: The metric requires correct allow/deny for all roles; a matrix prevents gaps and documents evidence for the quality report.

Dashboard filter response-time benchmark

Applies to: Dashboard Filter Response Time (R6).

Test description: In the documented test environment, apply typical filter changes (date range, region, granularity where present) and record time from user action until all visible KPI cards and charts show updated data (use browser performance tools or scripted timing). Repeat for a minimum sample size (for example 30 runs per filter type). Report 95th percentile latency and compare to ≤ 1000 ms.

Rationale: The requirement and metric are time-bound; percentile reporting matches the stated acceptance criterion and avoids flaky single-shot timings.

Dashboard filter stability and error-free completion run

Applies to: Dashboard Load Completion Rate (R6).

Test description: During the same filter benchmark session (or a dedicated soak), perform a long sequence of filter changes and tab or viewport switches (Fire versus EMS where implemented). Log client console errors, failed network requests, and blank charts. Compute the percentage of filter changes that complete without error; target at least 99% over the run.

Rationale: This metric separates functional completion from raw speed: a fast but error-prone dashboard would fail here even if occasional fast runs occurred.

Protected route and deep-link enforcement suite

Applies to: Protected Route Enforcement Rate (R7).

Test description: With no valid session, open protected URLs directly (e.g. `/home`, bookmarks). Confirm no protected content is shown: redirect to login (`/`) or `/session-expired` when a stale session is simulated. Log blocked vs incorrectly allowed.

Rationale: Measures unauthenticated access prevention, including bypass via direct links.

Authenticated login-to-home success run

Applies to: Login Session Success Rate (R7).

Test description: Sign in with valid test accounts that already satisfy profile + verification; expect `/home` without errors. Exclude accounts meant to stop at complete-profile or awaiting-access. Target $\geq 99\%$ success for these cases.

Rationale: Isolates successful session establishment for users allowed to use the app.

Profile completion redirect enforcement test

Applies to: Profile Completion Enforcement Rate (R8).

Test description: Use users missing required fields (e.g. pending_* username, missing phone, per your needsProfileCompletion rules). Navigate to /home; confirm redirect to /complete-profile before dashboard. Count redirects vs leaks.

Rationale: Matches the gate that blocks incomplete profiles from the dashboard.

Profile update completion success test

Applies to: Profile Update Completion Success Rate (R8).

Test description: From complete-profile, submit valid required fields; confirm persistence and navigation to /home without bounce-back. Include invalid submissions as non-success. Target ≥95% success on valid submissions.

Rationale: Verifies users can clear the gate with valid data, not only that they are redirected.

Unverified-account dashboard block test

Applies to: Verification Gate Enforcement Rate (R9).

Test description: Use verified=false (or equivalent) accounts with complete profiles. Attempt /home and protected flows; expect /awaiting-access and no KPI/upload access; add API checks if applicable. Target 100% block.

Rationale: Directly tests the verification gate on protected functionality.

Post-approval access regression test

Applies to: Post-Approval Access Success Rate (R9).

Test description: After admin approves/verifies test accounts, sign in and confirm /home and at least one KPI load works without extra manual fixes. Target ≥95%.

Rationale: Covers the approved-user end state of the verification lifecycle.

Admin user management scenario matrix

Applies to: Admin User Management Success Rate (R10).

Test description: As admin: list users, approve unverified user, change role, remove/disable if supported. Confirm UI, refetched list, and no server errors. Target ≥95% across scenarios.

Rationale: Matches “admin actions complete successfully” in the metric.

Admin console permission matrix

Applies to: Admin Console Permission Accuracy (R10).

Test description: For viewer, analyst, admin: try admin UI and admin APIs; expect only admin succeeds. Target 100% correct allow/deny.

Rationale: Separates access control accuracy from “management succeeds once you’re in.”